

Qi Xu

PhD Student in Artificial Intelligence @ Westlake University

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EDUCATION



Westlake University

PhD Student in Artificial Intelligence

Supervised by [Peidong Liu](#); Research: Embodied AI, Multimodal Large Language Models

2026 – Present



Wuhan University

Master in Pattern Recognition and Intelligent Systems

Research: Computer Vision, 3D Reconstruction, Multimodal Learning

2023 – 2026



Wuhan University

B.S. in Spatial Information and Digital Technology

2019 – 2023

RESEARCH INTERESTS

Multimodal Large Language Models (MLLMs), with emphasis on video reasoning and spatio-temporal grounding. Worked closely with [Dr. Xiangtai Li](#).

EXPERIENCE



ByteDance, TikTok Global Live LLM Strategy Department

2025 – 2026

Research Intern; Mentored by [Xiangtai Li](#)

- Developed and optimized **Multi-Modal Large Language Models (MLLMs)** for temporal grounding tasks using SFT and Reinforcement Learning (RL) pipelines.
- Built scalable data pipelines for massive MLLM training, enabling efficient collection and annotation of high-quality temporal grounding datasets.
- Conducted **large-scale distributed training on hundreds of GPUs**, optimizing training efficiency and model convergence for production-level deployment.
- Designed agentic workflows leveraging LLMs to automate real-time interaction analysis and optimize content distribution within the TikTok Live ecosystem.

SELECTED PUBLICATIONS



SIU3R: Simultaneous Scene Understanding and 3D Reconstruction Beyond Feature Alignment

[NeurIPS 25 Spotlight](#) [163 GitHub Stars](#) [12 Citations](#) [Project Page](#) [ArXiv](#) [Code](#)

Qi Xu*, Dongxu Wei*, Lingzhe Zhao, Wenpu Li, Zhangchi Huang, Shunping Ji, Peidong Liu

Introduced an alignment-free framework for simultaneous 3D reconstruction and scene understanding from unposed images via a shared pixel-aligned 3D representation.

My Contributions: Project Lead, Core Idea Contributor, and primary code implementer; led benchmarking, ablation analysis, and paper writing.

2025



Towards One-to-Many Temporal Grounding

[ICML 26](#) [Project Page](#) [ArXiv](#)

Qi Xu*, Yue Tan*, Shihao Chen, Jiahao Meng, Anran Wang, Shunping Ji, Hao Fei, Xiangtai Li

Introduced the first systematic solution for OMTG, featuring a 56k-sample dataset and RL training with CoT-based rewards; outperformed Gemini 2.5 Pro by 15.85%.

My Contributions: Core Idea Contributor, code implementer, and data pipeline constructor; led data construction, SFT/RL training pipeline implementation, reward design, model training, benchmarking, and paper writing.

2026



VideoZeroBench: Probing the Limits of Video MLLMs with Spatio-Temporal Evidence Verification

[arXiv 26](#) [Project Page](#) [ArXiv](#) [Code](#)

Jiahao Meng, Tan Yue, **Qi Xu**, Haochen Wang, Zhongwei Ren, Weisong Liu, Yuhao Wang, Renrui Zhang, Yunhai Tong, Haodong Duan

A hierarchical long-video QA benchmark that verifies model answers with temporal intervals and spatial boxes under a five-level evidence protocol.

My Contributions: Co-designed the five-level spatio-temporal evidence evaluation protocol; designed and implemented the annotation tool and guidelines; constructed approximately 200 complete QA-evidence samples.

2026



Watch, Remember, Reason: Human-View Video Understanding with MLLMs

[arXiv 26](#) [ArXiv](#) [GitHub](#)

Jiahao Meng, Yue Tan, **Qi Xu**, Kuan Gao, Weisong Liu, Yanwei Li, Xiangtai Li, Lingdong Kong, Haochen Wang, Qianyu Zhou, Jiangning Zhang, Guangliang Cheng, Yunhai Tong, Lu Qi, Minghsuan Yang

Surveyed human-view video understanding with MLLMs through watching, remembering, and reasoning abilities for long, multimodal, and knowledge-intensive video scenarios.

My Contributions: organized literature for the Memory section, created figures and tables, and contributed to paper writing and revision.

2026



HiCI: Hierarchical Construction-Integration for Long-Context Attention

ICML 26 [ArXiv](#) [Code](#)

Xiangyu Zeng, **Qi Xu**, Yunke Wang, Chang Xu

Efficient long-context modeling using hierarchical attention, extending LLaMA-2 to 100K tokens with only 5.5% additional parameters.

My Contributions: participated in partial method design and paper revision.

2026



Any 3D Scene is Worth 1K Tokens: 3D-Grounded Representation for Scene Generation at Scale

arXiv 26 [Project Page](#) [ArXiv](#) [Code](#)

Dongxu Wei*, **Qi Xu***, Zhiqi Li, Hangning Zhou, Cong Qiu, Hailong Qin, Mu Yang, Zhaopeng Cui, Peidong Liu

First approach to perform 3D scene generation directly within an implicit 3D latent space using 3D Diffusion Transformers.

My Contributions: completed large-scale dataset construction, ablation studies, evaluation, and contributed to paper writing and revision.

2026



E-MoFlow: Learning Egomotion and Optical Flow from Event Data via Implicit Regularization

NeurIPS 25 [Project Page](#) [ArXiv](#) [Code](#)

Wenpu Li, Bangyan Liao, Yi Zhou, **Qi Xu**, Pian Wan, Peidong Liu

An unsupervised framework for joint ego-motion and optical flow estimation using implicit neural representations and differential geometric constraints.

My Contributions: participated in partial method design and paper revision.

2025

SUPERVISED PUBLICATIONS



SupIR-GS: Thermal Infrared Super-Resolution Novel View Synthesis with Imaging-Calibrated 3DGS

ECCV 26 [Corresponding Author](#)

Jin Liu, Haodong Li, Jiagang Chen, Dabin leng, Jiguang Li, Zhao Huang, Xiaoshuai Zhang, **Qi Xu**[†], Zhiwen Zheng,

2026

Xingru Huang[†]

Reconstructs high-resolution 3D thermal scenes from low-resolution inputs via physics-informed degradation modeling.

Supervision: provided the core idea, guided method design and experiment planning, and led paper writing and revision.



Physically Grounded Dual-Opacity Gaussian Splatting for Joint RGB-TIR Reconstruction

ECCV 26 [Corresponding Author](#)

Jin Liu, Dabin leng, Jiagang Chen, Haodong Li, Jiguang Li, Zhao Huang, Xiaoshuai Zhang, Zhiwen Zheng, Xingru

2026

Huang[†], **Qi Xu**[†]

A unified RGB-TIR reconstruction framework resolving cross-spectral visibility conflicts through thermal field modeling.

Supervision: proposed the core idea, coordinated experiments and ablations, and led paper writing and revision.

SKILLS

Research: MLLM research on video reasoning, spatio-temporal grounding, SFT/RL post-training, video agents, benchmark construction, and test-time scaling.

Engineering: Python, PyTorch, distributed training/evaluation, data construction pipelines, and large-scale experimental analysis; familiar with C++ and Linux development.

Writing & Leadership: Leading research projects from problem formulation to implementation, benchmarking, analysis, paper writing, and public release.